Cloud Computing-II

Q.1 Answer the following.

(a) Short questions (3 marks)

(i) Which type of IP address is accessible over the internet: public or private?

A public IP address is accessible over the internet, whereas a private IP address is only accessible within a private network, such as a Virtual Private Cloud (VPC).

(ii) What cloud networking component determines how traffic is directed within and outside the VPC?

Routes are the cloud networking component that determines how traffic is directed within and outside the VPC.

(iii) What is meant by GKE?

GKE stands for Google Kubernetes Engine. It is a managed, production-ready environment for deploying containerized applications. It simplifies the process of managing, scaling, and orchestrating containers using Kubernetes.

(b) Objective type/MCQs/True-False/Fill in blanks (7 marks)

1. What is the main purpose of a Virtual Private Cloud (VPC) in cloud networking?

The main purpose of a Virtual Private Cloud (VPC) is to allow users to create isolated, customizable networks within a public cloud environment, providing control over network topology, IP address ranges, and security policies.

2. A Virtual Private Cloud (VPC) allows users to create isolated networks within a public cloud environment. True or False

True

3. What does PaaS stand for in the context of Google App Engine?

PaaS stands for Platform as a Service.

4. In cloud networking, **firewall rules** are used to control the flow of traffic to and from resources based on specified rules.
5. Google App Engine is a **Platform as a Service (PaaS)** that allows developers to build and host applications without managing infrastructure.
6. To deploy containerized applications in Google Cloud, developers commonly use **Google Kubernetes Engine (GKE)** for orchestration.

7. What is the use of a load balancer?

A load balancer distributes incoming network traffic across a group of backend resources (like virtual machines, containers, or instances) to ensure high availability, scalability, and reliability of applications.

Q.2 Answer the following.

(a) Two Questions of 2 Marks

(i) Define Multiple VPC?

Multiple VPCs refers to a network design where an organization uses more than one Virtual Private Cloud (VPC) within the same cloud project or across different projects. This approach is often used to logically separate different environments (e.g., development, staging, and production), business units, or applications for better security, management, and control.

(ii) What is the use of Routes in Cloud Computing?

In cloud computing, routes define the path that network traffic takes from a source to a destination. They specify how packets are forwarded from one subnet to another within a VPC or to a destination outside of the VPC, such as the internet or another network via VPN. Routes are essential for enabling communication between different resources in a cloud network.

(b) Two Questions of 3 Marks

(i) How can you create the instances using GCP?

You can create instances (Virtual Machines) in Google Cloud Platform (GCP) using the following methods:

- **Google Cloud Console (Web UI):** The user-friendly graphical interface allows you to specify all the instance details (machine type, image, disk, network, etc.) through forms and menus.
- **gcloud CLI:** The command-line interface is ideal for automation and scripting. You use the gcloud compute instances create command with various flags to specify the instance's configuration.
- **APIs and Client Libraries:** For programmatic creation, developers can use the Compute Engine API or client libraries in their preferred language (e.g., Python, Java) to build custom applications that manage instances.

(ii) Analyze the role of load balancer and its types.

A load balancer plays a critical role in cloud architecture by distributing incoming traffic to multiple backend resources. Its primary functions include:

- **High Availability:** By directing traffic away from unhealthy instances, it ensures the application remains online.
- **Scalability:** It can handle an increase in traffic by automatically distributing the load to newly added instances.
- **Performance:** It prevents any single instance from becoming a bottleneck, ensuring faster response times.

The main types of load balancers in a cloud environment are:

- **HTTP(S) Load Balancing:** A global load balancer that distributes HTTP/S traffic to backends based on URLs or hostnames.
- **TCP/UDP Load Balancing:** A regional load balancer that handles TCP/UDP traffic at the network layer.
- **Internal Load Balancing:** A regional load balancer that distributes traffic to instances within a single VPC network, often used to create a tiered application architecture.

Q.3 Attempt any TWO.

(iv) Write the command for Authorize and Firewall on GCP.

The gcloud command for creating a firewall rule is:

gcloud compute firewall-rules create [FIREWALL_RULE_NAME] --network=[NETWORK_NAME] --allow=[PROTOCOL:PORT] --source-ranges=[IP_RANGE]

For example: gcloud compute firewall-rules create allow-http --network=default --allow=tcp:80 --source-ranges=0.0.0.0/0

(v) Analyse the services of Azure Cloud.

Microsoft Azure offers a vast range of services, including:

- **Compute:** Services like Azure Virtual Machines (IaaS) for hosting workloads, Azure Functions (FaaS) for serverless computing, and Azure Kubernetes Service (AKS) for container orchestration.
- **Storage:** Services such as Azure Blob Storage for unstructured data, Azure File Storage for file shares, and Azure Disk Storage for virtual machine disks.

- **Databases:** A wide selection of database services including Azure SQL Database, Azure Cosmos DB (a globally distributed multi-model database), and open-source database services.
- **Networking:** Services like Azure Virtual Network (VNet) for private networks, Azure Load Balancer, and Azure DNS.

(vi) Give me a command for VM instance creation in the name of "FITCS" and Zone is "Vadodara".

gcloud compute instances create FITCS --zone=Vadodara

(Note: 'Vadodara' is not a valid GCP zone. A valid zone would be something like 'asia-south1-a'. Assuming the question uses a conceptual name, the command structure is correct.)

Q.4 Answer the following.

(a) How can you host the web application using GCP with two pages of Frontend & Backend instances?

1. **Create VM Instances:** Create two separate VM instances in a VPC. One instance will host the frontend application, and the other will host the backend.
2. **Configure Firewall Rules:** Create a firewall rule to allow HTTP/S traffic (port 80/443) from the internet to the frontend instance's IP address. Create another rule to allow the frontend instance to communicate with the backend instance on the appropriate port.
3. **Deploy Applications:** Deploy the frontend application (HTML, CSS, JS) to the frontend instance and the backend application (e.g., Node.js, Python, Java) to the backend instance.
4. **Set Up Load Balancer (Optional but recommended):** For production, create an HTTP(S) Load Balancer that routes external traffic to the frontend instance, providing a single public IP address and distributing traffic if you use multiple frontend instances.

(b) Create the Cloud platform using Google Cloud and use the Cloud Bucket Storage in the name "MCA" and inside the directory save your image.

1. **Create a Project:** First, you need a GCP project. If you don't have one, create it via the Cloud Console or gcloud projects create.
2. **Create a Storage Bucket:** Use the gsutil command-line tool to create a bucket named "MCA".
    - gsutil mb gs://mca
3. **Upload an Image:** To save an image inside this bucket, use the gsutil cp command.

- gsutil cp /path/to/your/image.jpg gs://mca/

(Note: This saves the image directly in the root of the bucket. If you need a "directory" inside the bucket, the path would be gs://mca/directory-name/image.jpg.)